Algal Blooms in Lake Chapala: A Differential Equation Model

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July 2026

Abstract

Lake Chapala, Mexico's largest freshwater lake, experiences recurring algal blooms that affect drinking water quality for millions of residents. We develop a differential equation model for algae population density that combines periodic nutrient influx from agricultural pesticide cycles with proportional removal representing intervention and natural loss. Starting from a deliberately simple oscillatory growth term, we systematically refine the model through five versions to enforce biological feasibility: non-negative population, population-dependent growth, and proportional death. The resulting model: $dy/dt = a(1 - \cos(bt))y - hy$ is calibrated using published observations of spraying frequency (15 days), maximum growth rate (25%), and net decline (0.5% per day). We derive a closed-form solution via separation of variables and analyze when population density exceeds an odor threshold of 100 g/m^2. For initial density 81 g/m^2, the model predicts five early exceedance windows that shorten over time, with blooms remaining below threshold after day 72. The work illustrates a systematic approach to building interpretable environmental models from qualitative constraints and published data.

Keywords: algal blooms; Lake Chapala; population dynamics; differential equations; environmental modeling

1 Introduction

Lake Chapala in central Mexico is more than just a lake; it supports fishing and farming communities, provides drinking water to millions in Guadalajara, and hosts a diverse fish population and migratory birds. Because it sits inside the larger Lerma River system, Lake Chapala also serves as an environmental indicator – problems in Lake Chapala are indicators of the health of the entire ecosystem. For a long time, the lake has been going through dehydration caused by evaporation, increasing irrigation demands, drainage by the Santiago River, and water being piped to Guadalajara (a major nearby city). As the lake level drops, the water has also become "cloudy", which matters because it affects how light and nutrients move through the ecosystem.

One of the most visible and disruptive symptoms of these changes is recurring algal blooms. Even when blooms are not toxic, they still create real problems. They can be a nuisance for tourists and nearby communities, as reports describe blooms and associated plant growth (like seaweed and hyacinth) that interfere with motorized fishing boats [1]. Algal blooms have also been linked to unpleasant taste and odor in drinking water for people in Guadalajara, requiring additional treatment.

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Agricultural chemical residues from runoff, including heavy metals and dissolved solids, can drastically accelerate algae growth. However, algae growth is not controlled by this single cause alone. In addition to these residues, Lake Chapala has high levels of nutrients such as nitrogen and phosphorus and is close to being hyper-eutrophic, meaning nutrient levels are extremely high for the ecosystem. Early studies suggested that nitrogen levels and clay turbidity (how cloudy the water is) were major limiting factors for algae growth. In addition, clay particles from erosion and agricultural runoff increase both nutrients like nitrogen and phosphorus in the water and the water's turbidity, connecting land-use changes around the lake to increased algae growth.

With these interacting drivers in mind, the goal of this paper is to build a simple, interpretable differential equation model for algae population density over time and use it to understand why blooms can recur and persist even when interventions attempt to reduce algae levels.

2 Model Development

We begin model development with an attempt to develop a resource-dependent model to describe the resilient ability of an algae population to bounce back after near-annihilation. To construct such a realistic model for algal blooms in Lake Chapala, we begin with the general structure

$$\frac{dy}{dt} = \text{Growth Function} - \text{Death Function}. \tag{1}$$

Here, $y(t)$ represents the density of the algae population (g/m^2), and t represents time (days). Because the background suggests that chemical influx occurs periodically, we seek to model growth using an oscillating function over time.

2.1 Model: V1

To incorporate this periodic growth, the first model we consider is

$$\frac{dy}{dt} = -\cos(t). \tag{V1}$$

We plot this differential equation on a slope field plotter and observe the following characteristics of the general solution.

As we can see in the slope field (Figure 1), this model represents an oscillating rate of change of population over time, varying between -1 and $+1$. The cosine function was chosen to reflect periodic growth and decline due to recurring agricultural chemical input. The derivatives, $-\cos(t)$, oscillate between -1 and $+1$, since

$$-1 \leq -\cos(t) \leq 1.$$

However, this model poses several limitations and introduces scenarios that conflict with biological reality. To determine whether this oscillating growth rate also produces an oscillating population, we solve the differential equation model **V1**:

$$\frac{dy}{dt} = -\cos t,$$

$$dy = -\cos(t)\, dt.$$

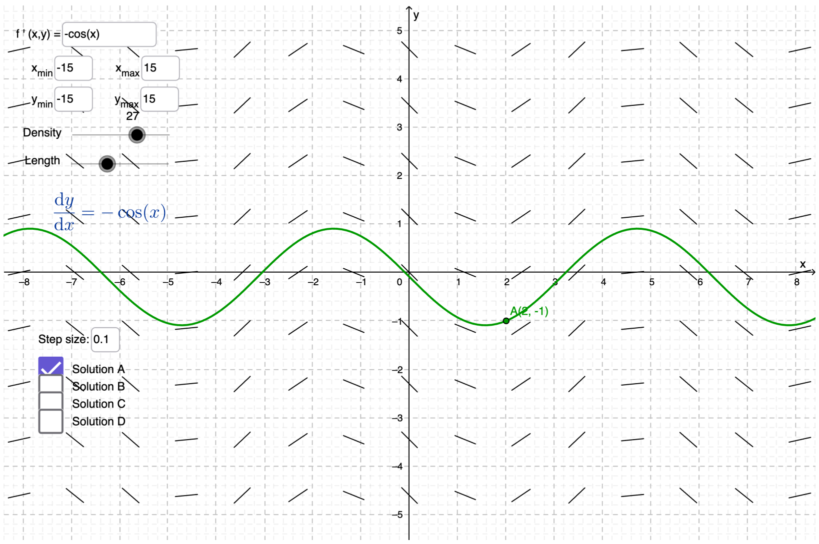

Figure 1: Slope field for Differential Equation Model **V1**, showing periodic oscillation in the rate of change.

Integrating, we get

$$y(t) = -\sin(t) + C.$$

Because $-\sin t$ oscillates between -1 and $+1$, the population exhibits oscillatory behavior. However, this model fails to address several important biological features and, in some cases, predicts behavior that is clearly not biologically feasible.

First, at $t = 0$,

$$\frac{dy}{dt} = -\cos(0) = -1,$$

so the model begins with negative growth.

Second, the derivative does not depend on the population y. This implies that the same amount of growth or decline occurs regardless of whether the algae population is large or small. Biologically, this is unrealistic.

Most importantly, when the population reaches zero, the derivative may still be negative. This would cause the solution to move to negative population values, which are not biologically meaningful. Thus, model V1 does not satisfy the requirement that the population remain nonnegative.

These limitations motivate us to modify the growth function to address the above concerns, ensure biological feasibility, and better represent realistic population dynamics.

2.2 Model: V2

To address the aforementioned limitations of model V1, particularly the issue of negative growth at zero population and the lack of feasibility, we modify the oscillating growth function by shifting it upward. The revised model now becomes

$$\frac{dy}{dt} = 1 - \cos(t) \tag{V2}$$

In this model, the cosine function is shifted upward by its amplitude, which is 1. Since $\cos t$ ranges between -1 and $+1$, the expression $1 - \cos t$ now ranges between 0 and 2. As a result, the rate of change is never negative.

This upward shift is helpful because it prevents the model from forcing the population to decrease when the population is already small. As observed in model V1, the derivatives oscillated between -1 and $+1$, meaning the population could be pushed into negative values. In model V2, however, the derivative is always nonnegative.

Below, we plot the differential equation model V2 on a slope field plotter and examine its features.

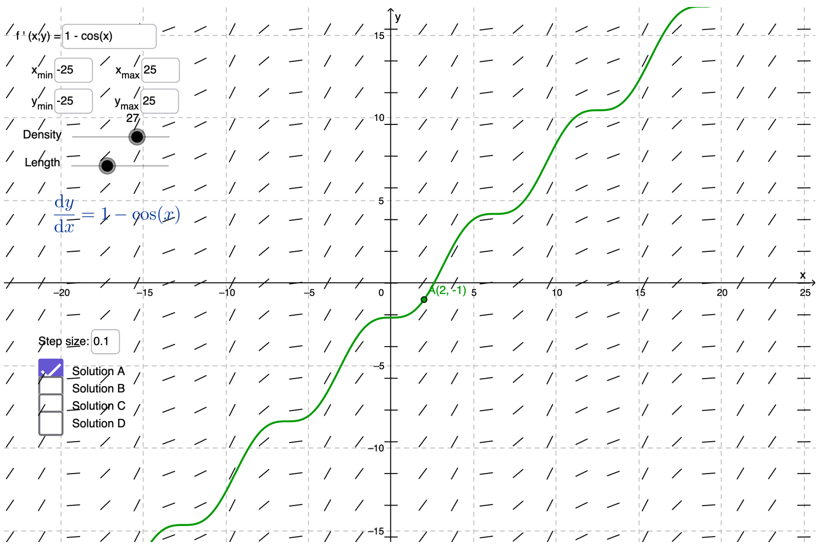

Figure 2: Slope field for model V2, showing the upward-shifted oscillating rate of change.

At $t = 0$,

$$\frac{dy}{dt} = 1 - \cos(0) = 0,$$

so the model begins with zero growth rather than negative growth, which resolves the negative population issue. Thus, shifting the oscillation upward ensures that growth remains biologically feasible and prevents the derivative from driving the population below zero.

However, although this modification improves feasibility, the derivative still does not depend on the population y. This means the same amount of algae is added to the total population due to growth regardless of how large or small the population is, which remains biologically unrealistic. Therefore, model V2 still lacks a mechanism that ties growth to the current population size.

2.3 Model: V3

Since model V2 treated growth as a fixed amount added to the population, independent of the current population size, it remained biologically unrealistic. In the real world, population growth typically depends on how large the population already is. Therefore, to introduce dependency on the population, we modify the oscillating term from the previous model so that it acts as a growth rate rather than a fixed growth amount. The revised model is

$$\frac{dy}{dt} = (1 - \cos(t))y \tag{V3}$$

In this model, the oscillating term $1 - \cos(t)$ multiplies the population y, making the growth both time- and population-dependent. This means that when the population is larger, the actual amount of growth is larger, and when the population is small, the amount of growth is proportionally smaller, which better reflects how growth occurs in real biological systems.

We plot the differential equation model V3 on a slope field plotter below and observe how population y changes over time t.

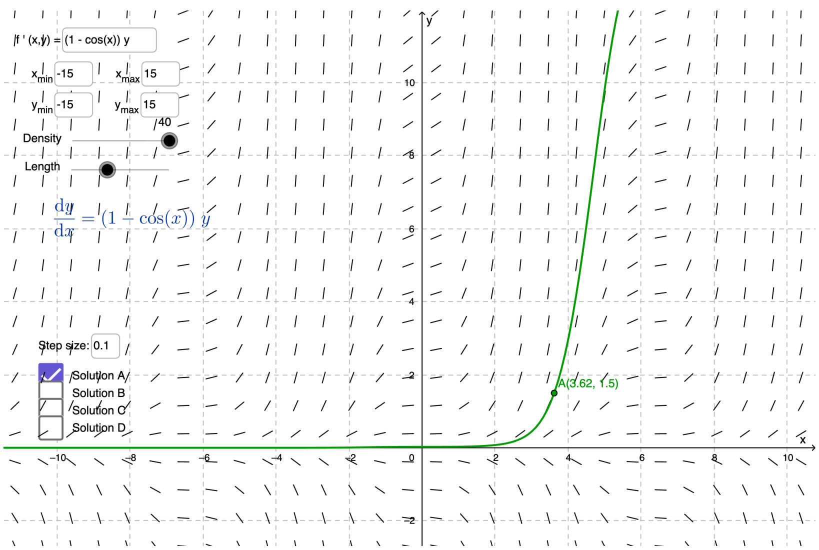

Figure 3: Slope field for model V3, showing slopes that depend on both time and population size.

As observed in the slope field (Figure 3), model V3 differs from model V2 because the slopes now depend on both time and population size. In Version 2, slopes varied with time but remained the same vertically for all values of y. However, in Version 3, the slopes become steeper as y increases and flatten near $y = 0$. This indicates that the population increases by a fraction of itself rather than by a fixed number. Thus, when the population is larger, it grows more rapidly, and when the population is smaller, it grows more slowly.

Biologically, this shows the difference between adding a constant number of algae each day (actual growth) and allowing the algae to grow at a fraction of its current total population (rate of growth). This model resolves several issues from previous models. However, it still lacks a way to represent removal or governmental intervention. Therefore, further refinement of the model remains necessary.

2.4 Model: V4

Although model V3 incorporates growth more realistically by making it proportional to the population, it does not yet account for external intervention or removal; the death function is still zero. To represent consistent governmental efforts to reduce algae growth, we introduce a constant death term. The revised model is

$$\frac{dy}{dt} = (1 - \cos(t))y - 1 \tag{V4}$$

Here, the newly introduced term -1 represents a constant removal of algae over time, independent of the current population size. Visually, this refined model preserves the periodic behavior of the population function while incorporating a constant death term. Compared to model V3, the population still grows periodically, but each cycle is reduced by a fixed amount due to the constant death function.

Below, we plot model V4 on a slope field plotter to observe the changes due to introducing the death function.

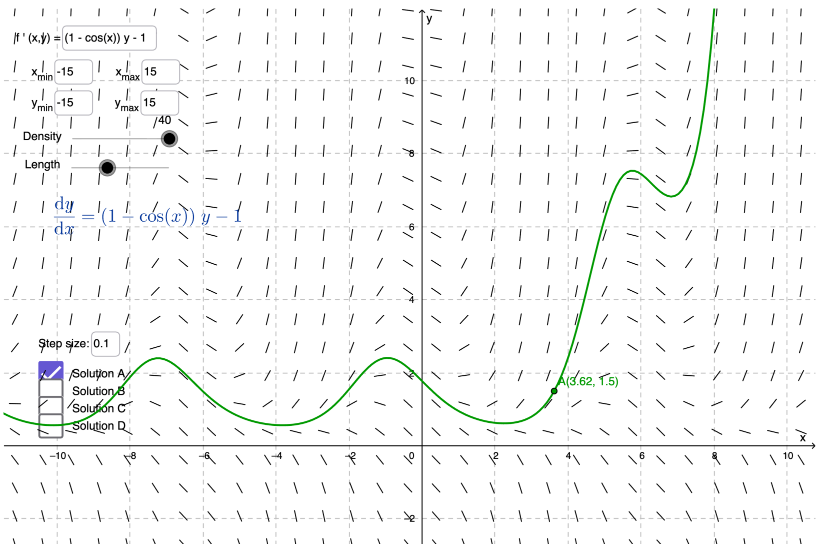

Figure 4: Slope field for model V4, illustrating periodic growth with constant removal.

However, this assumption is also not biologically realistic. If the population becomes very small, removing the same fixed amount could drive the population below zero. In fact, if y is sufficiently small, the constant death term may outgrow the growth term, forcing the solution into negative values near $y = 0$.

Since population cannot be negative, this model may again violate feasibility. Therefore, although model V4 introduces an important biological feature—government intervention—it does so in a way that is not entirely realistic. It is not feasible and often not realistic to assume that equal numbers of algae are continuously removed regardless of the population size. This leads us to modify the death function so that it becomes proportional to the population itself.

2.5 Model: V5

As discussed in model V4, removing a fixed amount of algae at all times is not biologically realistic. To address this limitation, we modify the death term so that removal becomes proportional to the current population. With that in mind, the revised model we propose is

$$\frac{dy}{dt} = (1 - \cos(t))y - y \tag{V5}$$

In this refined model V5, the death term is now proportional to the population size. This means that when the population is large, more algae are removed, and when the population is small, less algae are removed. Biologically, this is more reasonable, since the death function due to intervention efforts or natural losses is often proportional to the current total population rather than fixed at a constant amount.

Visually, this modification prevents the solution from being forced below zero. If the population reaches zero, then

$$\frac{dy}{dt} = 0.$$

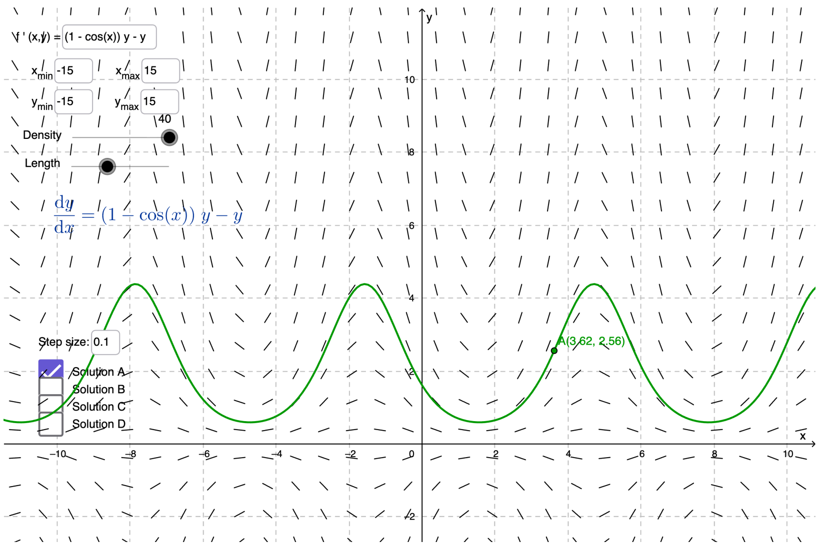

Figure 5: Solution curve for model V5, preventing negative population even when population is near zero.

So, the solution remains at zero and does not suddenly become negative. Therefore, if the initial population is nonnegative, the solution remains nonnegative for all time. This ensures biological feasibility.

Looking back at model V3, where the population grew without bound, the addition of a proportional death function creates more realistic population dynamics, allowing the population to fluctuate in magnitude while remaining feasible. Version 5 therefore provides a more realistic balance between the periodic growth function and the death function, and leads us very close to the generalized model.

2.6 General Model

After refining the model through V1–V5, we now introduce a general form that incorporates adjustable parameters to provide flexibility and better represent the behavior of algal bloom in Lake Chapala. The general model is

$$\frac{dy}{dt} = a(1 - \cos(bt))y - hy \tag{2}$$

This model combines periodic growth with a proportional death term. The term $a(1 - \cos bt)y$ represents time- and population-density-dependent growth, and the term $-hy$ represents proportional death due to government intervention and natural loss.

If time t is measured in days and population density y is measured in grams per square meter (g/m^2), then the derivative $\frac{dy}{dt}$ has units of $g/(m^2 \cdot \text{day})$. We now describe what each parameter represents and determine its units.

Parameter a: The parameter a controls the strength of the periodic growth. Larger values of a increase the overall growth rate due to chemical influx. Since this term multiplies y, the parameter a has units of 1/day.

Parameter b: The parameter b controls how frequently the oscillation occurs. Since the cosine function repeats every 2π, b determines how fast the cycles repeat or, in other words, the period of the growth cycles. The parameter b must have units of radians per day, rad/day, to ensure the expression bt inside the cosine function is dimensionless.

Parameter h: The parameter h represents the proportional death rate, which accounts for consistent removal of algae over time. Like the parameter a, this parameter also has units of 1/day.

This general model provides flexibility in tuning the amplitude of growth, the frequency of oscillation, and the proportion of removal through the associated parameters. It therefore helps us fit the model to observed data and analyze long-term algae population behavior in Lake Chapala.

3 Parameter Calibration

We determine suitable values for the parameters a, b, and h from the general model 2 using the following observed data. The goal is to select parameter values that reflect the observed spraying frequency, maximum rate of change, and subsequent decrease in growth over time.

The observed experimental data follows:

1. Agricultural pesticides are sprayed approximately every 15 days [1].
2. The maximum recorded rate of change of the algae population was 25% [1].
3. The maximum rates of change diminish slowly over time at approximately 0.5% per day [1].

Parameter b: Since spraying occurs every 15 days, the growth cycle has a period $T = 15$ days. We also know that the cosine function repeats after every 2π radians. Therefore, for one full cycle to occur in our model, the input of the cosine function increases by 2π after one period T.

In the expression $\cos(bt)$, the input is bt. Thus, one full cycle occurs when

$$b(t+T) - bt = 2\pi$$

or,

$$bt + bT - bt = 2\pi,$$

or,

$$bT = 2\pi.$$

Substituting $T = 15$ days,

$$b \cdot 15 = 2\pi.$$

So,

$$b = \frac{2\pi}{15}.$$

Thus, $b = \frac{2\pi}{15}$ radians per day.

Parameters a and h: The general model 2 is

$$\frac{dy}{dt} = a(1 - \cos(bt))y - hy. \tag{2}$$

Since the maximum percent rate of change is 25%, or 0.25 per day, and the maximum rate of change occurs when $1 - \cos(bt)$ is largest, which is 2, therefore, at its maximum,

$$0.25 = 2a - h.$$

On the other hand, the maximum rates diminish at approximately 0.5% per day. Therefore, the overall net growth rate is

$$a - h = -0.005.$$

Solving for a and h, we get

$$a = 0.255$$

and

$$h = 0.260.$$

Another way to interpret the statement that "the maximum percent rate of change is 25%" is to assume it refers to the maximum amplitude of the population density function. Therefore, we instead have

$$a' = 0.25$$

and, since

$$a' - h' = -0.005,$$

so

$$h' = 0.255.$$

4 Analytical Solution

We solve the differential equation model 2 to obtain $y(t)$, the algae population density as a function of time t.

4.1 General solution

Assuming $h \neq a$ and applying the initial condition $y(0) = y_0$, we solve the general model. Since this differential equation is linear in y (and therefore separable), we solve it using separation of variables.

The model:

$$\frac{dy}{dt} = a(1 - \cos(bt))y - hy$$

factoring out y,

$$\frac{dy}{dt} = y[a(1 - \cos(bt)) - h]$$

separating variables,

$$\frac{1}{y}\, dy = [a - h - a\cos(bt)]\, dt.$$

Integrating both sides,

$$\int \frac{1}{y}\, dy = \int (a - h)\, dt - \int a\cos(bt)\, dt,$$

or,

$$\int \frac{1}{y}\, dy = \int (a-h)\, dt - \int a\cos(bt)\, dt,$$

or,

$$\ln y = (a-h)t - \frac{a}{b}\sin(bt) + C.$$

Exponentiating,

$$y(t) = e^{(a-h)t - \frac{a}{b}\sin(bt) + C}.$$

or,

$$y(t) = e^{(a-h)t - \frac{a}{b}\sin(bt)} \cdot e^{C}.$$

or,

$$y(t) = C\, e^{(a-h)t - \frac{a}{b}\sin(bt)}. \tag{3}$$

4.2 Particular solution

Plugging in the initial condition $y(0) = y_0$ in the general solution (3), we get $C = y_0$. Therefore, the solution is

$$y(t) = y_0\, e^{(a-h)t - \frac{a}{b}\sin(bt)}. \tag{4}$$

5 Threshold Analysis

Substituting the parameter values computed in Section 3 and the initial population density,

$$y(0) = 81\, g/m^2,$$

into (4), we write the two particular solutions:

Using $a = 0.255$ and $h = 0.260$,

$$y(t) = 81\, e^{-0.005t - \frac{3.825}{2\pi}\sin(\frac{2\pi}{15}t)} \tag{5}$$

and

Using $a' = 0.25$ and $h' = 0.255$,

$$y'(t) = 81\, e^{-0.005t - \frac{3.75}{2\pi}\sin(\frac{2\pi}{15}t)}. \tag{5'}$$

Since both functions (5) and (5') generate nearly identical graphs while preserving all desired qualitative features of the model, it suffices to graph only one. Therefore, without loss of generality, we graph $y'(t)$ below.

Assuming population densities above 100 g/m^2 trigger offensive odors in the drinking water, we estimate the length of time intervals during which

$$y'(t) > 100.$$

1. 1st interval: $[9, 14]$, (length = 5 days)
2. 2nd interval: $[24, 29]$, (length = 5 days)

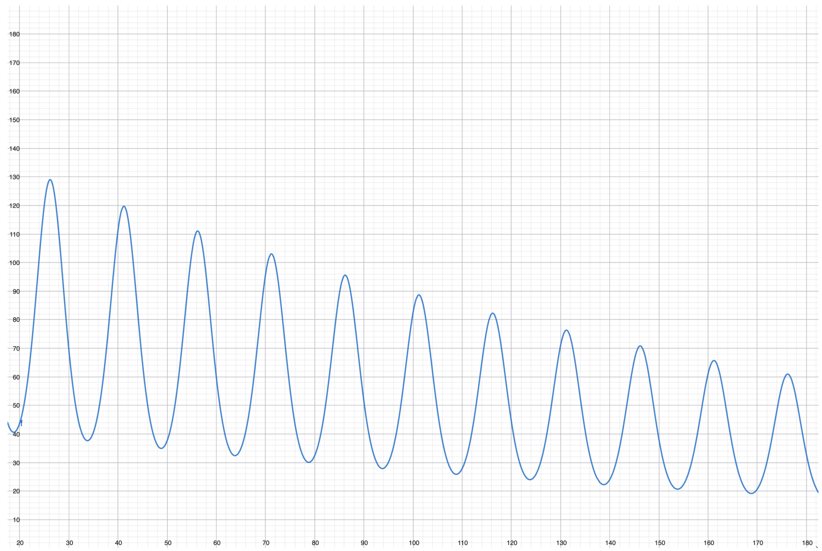

Figure 6: Graph of the particular solution $y'(t)$ with initial population density $y(0) = 81$ g/m^2 and calibrated parameter values.

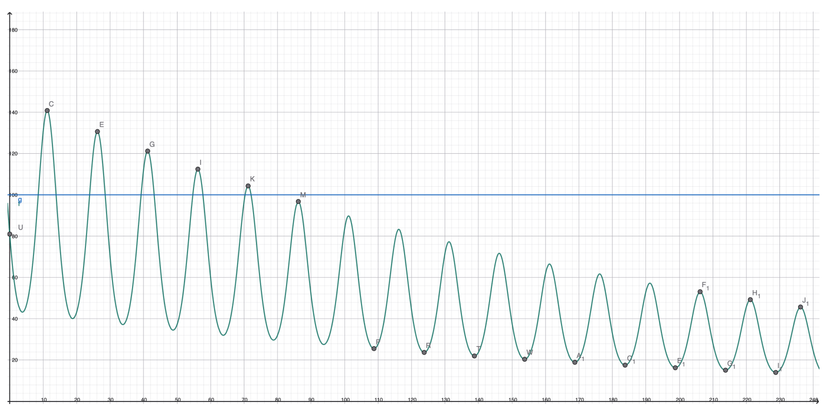

Figure 7: Graph of the particular solution $y'(t)$ with population density threshold $y(t) = 100$ g/m^2.

3. 3rd interval: $[39, 43]$, (length $= 4$ days)
4. 4th interval: $[55, 58]$, (length $= 3$ days)
5. 5th interval: $[70, 72]$, (length $= 2$ days)

We observe that the exceeding intervals become progressively shorter over time due to the negative net growth, $a - h = -0.005$. The last time the population exceeds 100 g/m^2 is approximately at $t = 72$ days. After that, even the peak amplitude stays below the odor threshold.

6 Conclusion

Throughout the paper, we developed a differential equation model to describe algal bloom behavior in Lake Chapala. Starting from a simple oscillatory growth idea, we systematically refined the model to ensure biological feasibility and realistic population behavior. By incorporating periodic growth and proportional death as suggested in the reference paper [1], we structured a general model that reflects both seasonal chemical influx (growth function) and consistent removal effects (death function).

Using the observed experimental data, we calibrated the parameters to reflect spraying frequency, maximum rate of change, and subsequent decrease in growth over time. We solved the model and analyzed how the algae population evolves over time. The solution shows oscillatory behavior with decreasing amplitude, meaning the blooms become less severe over time due to the negative net growth rate.

With initial population density of 81 g/m^2, the model predicts that the algae population exceeds the odor threshold of 100 g/m^2 during five early time intervals. However, these intervals shorten progressively, and the final exceedance occurs around day 72. After this point, the population remains below the threshold.

Overall, this mathematical model serves as a reasonable and simplified representation of Lake Chapala and, more specifically, the algal bloom behavior in this lake. It demonstrates how differential equations can be used to model environmental systems and interpret real-world data.

Acknowledgments

This research was conducted under the supervision of Professor Cayce Fylling (Gettysburg College). Environmental observations and scenario data are drawn from Harwood [1]. Code and supplementary materials are available at `https://github.com/soikat27/algal-bloom-de-model`.

References

[1] Richard Corban Harwood. Algal blooms threatening lake chapala. SIMIODE Modeling Scenario, 2015. Accessed July 2026. URL: `https://www.simiode.org/resources/2304`.